

Log Analytics in GCP

Log Analytics in Google Cloud Platform (GCP) refers to the process of collecting, storing, analyzing, and visualizing log data generated by cloud resources, applications, and services running on GCP. This is primarily facilitated through **Cloud Logging** (formerly Stackdriver Logging) and integrated with **Cloud Logging Query Language**, **BigQuery**, **Cloud Monitoring**, and **Cloud Operations suite**.

Key Components of Log Analytics in GCP

- 1. Cloud Logging**
Centralized logging service that ingests log data from GCP services, system components, and applications.
- 2. Log Router**
Filters and routes logs to supported destinations such as Cloud Storage, BigQuery, or Pub/Sub for further analysis.
- 3. Cloud Logging Query Language (LQL)**
Enables structured querying of logs using a powerful query syntax.
- 4. Log-based Metrics**
Allows creation of custom metrics based on log entries to trigger alerts and monitoring dashboards.
- 5. Integration with BigQuery**
Enables advanced analytics and machine learning workflows on log data using SQL.

Use Cases

- **Security and Compliance Monitoring**
Analyze access logs to detect anomalies, unauthorized access, and support audit requirements.
- **Application Performance Monitoring**
Correlate logs with metrics to diagnose performance bottlenecks or application failures.
- **Infrastructure Monitoring**
Monitor the behavior and health of virtual machines, Kubernetes clusters (GKE), and network components.
- **Incident Response and Root Cause Analysis**
Use logs to investigate incidents and trace failures back to their source.
- **Cost Optimization**
Analyze logs to identify underutilized resources or inefficient application behaviors.
- **Custom Alerts and Notifications**
Set up log-based metrics that trigger alerts when specific patterns or thresholds are detected.

Key Benefits

- **Fully Managed Service**
No infrastructure overhead; scalable and reliable logging infrastructure managed by Google.
- **Real-time Log Ingestion**
Near real-time log processing enables timely alerts and responses.
- **Seamless Integration with GCP Services**
Native support for all major GCP services including Compute Engine, Cloud Run, GKE, and App Engine.
- **Granular Access Control**
IAM-based access control ensures secure log access and management.
- **Compliance and Retention Support**
Configurable log retention policies help meet regulatory requirements.
- **Flexible Export Options**
Export logs to BigQuery for advanced analysis, to Cloud Storage for long-term retention, or to Pub/Sub for stream processing.

Key Features

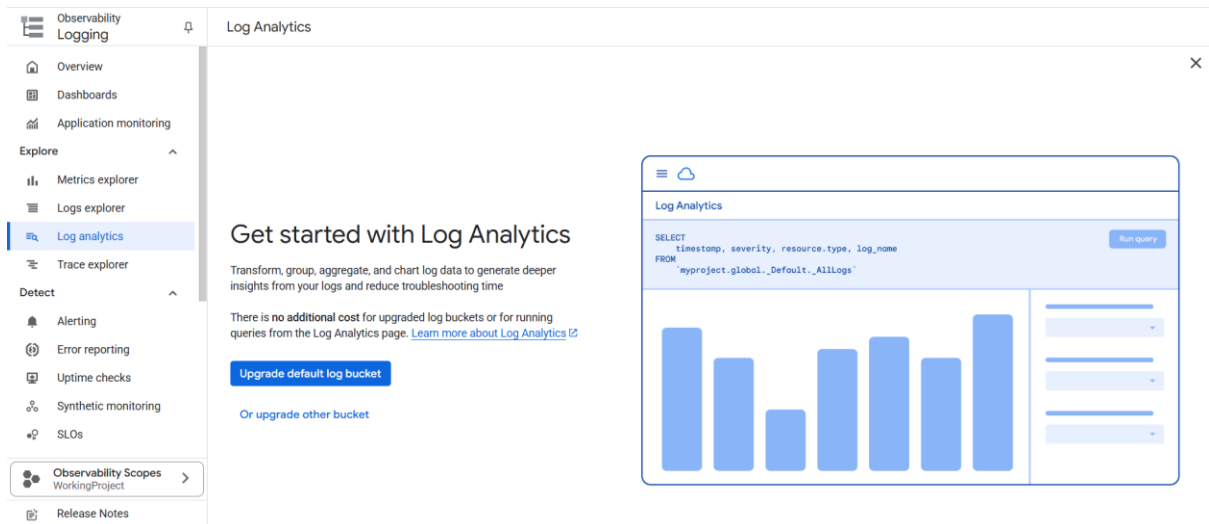
- **Log Viewer:** GUI for browsing and querying logs in Cloud Console.
- **Advanced Filtering:** Query logs using structured fields with the Logging Query Language.
- **Custom Log-based Metrics:** Create metrics based on log patterns for alerting and dashboarding.
- **Audit Logs:** Automatically captures Admin Activity, Data Access, System Events, and Policy Denied logs.
- **Structured and Unstructured Logging:** Supports JSON-formatted structured logs and plain text logs.
- **Integration with Monitoring and Alerting:** Build dashboards and alerts that combine log-based and metric-based insights.

Summary

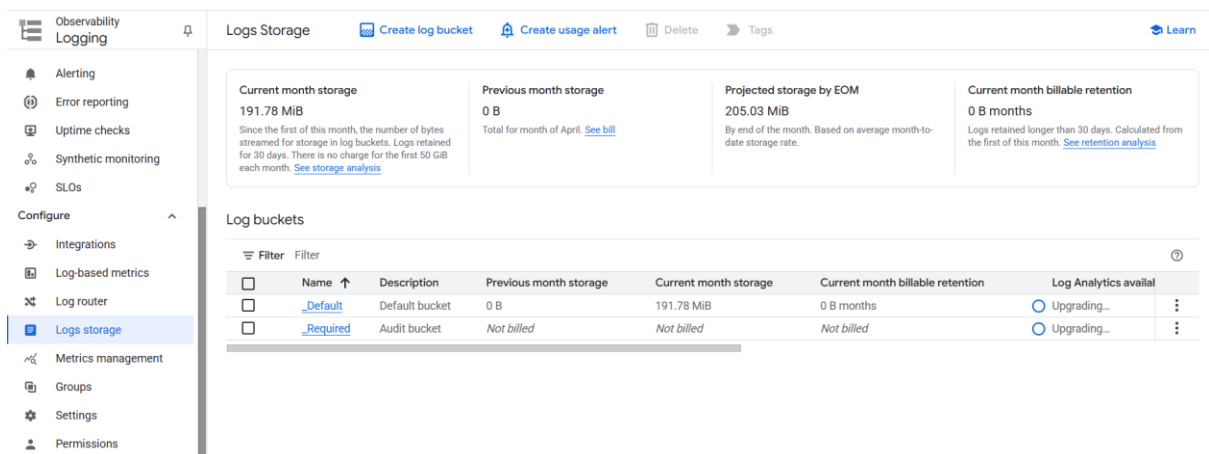
Log Analytics in GCP enables organizations to gain deep operational, security, and business insights from their log data. It provides a comprehensive, integrated, and scalable solution for managing logs across all GCP environments, facilitating better observability, faster issue resolution, and proactive infrastructure management.

To begin with the Lab

1. Search Log Analytics in GCP and go to it. In its dashboard, you will see that to get started, you need to upgrade your default log bucket. Click on Upgrade.



- Then search for Log Storage, here you will see two buckets (Default and Required) you need to upgrade both of them. We have upgraded our Default bucket but we also need to upgrade the Required.
- So, just click on upgrade and you will see both of them in upgrading state.



- Here you can see that both of your buckets have been upgraded.

Log buckets

Filter	Filter						
☐	Name	Description	Previous month storage	Current month storage	Current month billable retention	Log Analytics availal	
☐	_Default	Default bucket	0 B	191.78 MiB	0 B months	Open Upgraded	
☐	_Required	Audit bucket	Not billed	Not billed	Not billed	Open Upgraded	

- Now, on the Log Analytics you will see the dashboard like this. Here you can queries and get the real time logs.

- Come back to the Log buckets and click on Open on the Required bucket, then choose All logs. Now log analytics will load for this bucket.

Log buckets

Filter	Name	Description	Previous month storage	Current month storage	Current month billable retention	Log Analytics available
☐	_Default	Default bucket	0 B	191.78 MiB	0 B months	☒ Open Upgraded
☐	_Required	Audit bucket	Not billed	Not billed	Not billed	☒ Open Select a view _AllLogs

- On the Log Analytics in the query window, you will see a SQL based query. You can also change the time based on your preference.
- Click on Run query to see the logs. The logs are based on the services you are using or have used, but if you don't have anything running, then you will see no logs.
- If that happens, then you can open the logs for the Default bucket.

- In the logs here shown in the snapshot you will see that they are related to an instance.

Log Analytics

Query Recent (2) Saved (0)

Format Clear SQL reference

Ready to run

```

1 SELECT
2   timestamp, severity, resource.type, log_name, text_payload, proto_payload, json_payload
3 FROM
4   `still-kit-459403-e2.global._Default._Default`
5 LIMIT 10000

```

Results (5) Table Chart Both

Download Create alert Save to dashboard

Row	timestamp TIMESTAMP	severity STRING	type STRING	log_name STRING	text_payload STRING	proto_payload RECORD	js
1	2025-05-30 12:34:39.467 IST	DEBUG	gce_instance	projects/still-kit-459403-e2/logs/ops-agent-health	Null	Empty	
2	2025-05-30 12:35:55.320 IST	DEFAULT	gce_instance	projects/still-kit-459403-e2/logs/syslog	Null	Empty	
3	2025-05-30 12:35:55.338 IST	DEFAULT	gce_instance	projects/still-kit-459403-e2/logs/syslog	Null	Empty	
4	2025-05-30 12:35:55.338 IST	DEFAULT	gce_instance	projects/still-kit-459403-e2/logs/syslog	Null	Empty	
5	2025-05-30 12:35:55.335 IST	DEFAULT	gce_instance	projects/still-kit-459403-e2/logs/syslog	Null	Empty	

11. If you look at the query carefully, you will see that there you so many fields that we are selecting. We can also remove the fields that are NULL or not needed.

12. In the query, we removed the fields that were null and we got the data accordingly.

Log Analytics

Query Recent (3) Saved (0)

Format Clear SQL reference

Ready to run

```

1 SELECT
2   timestamp, severity, resource.type, log_name, json_payload
3 FROM
4   `still-kit-459403-e2.global._Default._Default`
5 LIMIT 10000

```

Results (5) Table Chart Both

Download Create alert Save to dashboard

Row	timestamp TIMESTAMP	severity STRING	type STRING	log_name STRING	json_payload JSON
1	2025-05-30 12:34:39.467 IST	DEBUG	gce_instance	projects/still-kit-459403-e2/logs/ops-agent-health	{code: "LogPingOpsAgent"}
2	2025-05-30 12:35:55.320 IST	DEFAULT	gce_instance	projects/still-kit-459403-e2/logs/syslog	{message: "2025-05-30T07:05:"}
3	2025-05-30 12:35:55.338 IST	DEFAULT	gce_instance	projects/still-kit-459403-e2/logs/syslog	{message: "2025-05-30T07:05:"}
4	2025-05-30 12:35:55.338 IST	DEFAULT	gce_instance	projects/still-kit-459403-e2/logs/syslog	{message: "2025-05-30T07:05:"}
5	2025-05-30 12:35:55.335 IST	DEFAULT	gce_instance	projects/still-kit-459403-e2/logs/syslog	{message: "2025-05-30T07:05:"}

13. Now run the query given below, in this query we are saying that please count the resources based on their types.

```

SELECT
  resource.type, COUNT(*) as count
FROM
  `still-kit-459403-e2.global._Default._Default`
GROUP BY resource.type

```

14. Here you can see that we can only see one service that is Virtual Machine.

Log Analytics

Query Recent (4) Saved (0)

Format Clear SQL reference

Ready to run

```

1 SELECT
2   resource.type, COUNT(*) as count
3 FROM
4   `still-kit-459403-e2.global._Default._Default`
5 GROUP BY resource.type

```

Results (1) Table Chart Both

Download Create alert Save to dashboard

Row	type	count
	STRING	INTEGER
1	gce_instance	10

15. You can also view the data in the form of a table, a chart or in both ways. Based on the services you can change the values to display your chart.

Log Analytics

Query Recent (4) Saved (0)

Format Clear SQL reference

Ready to run

```

1 SELECT
2   resource.type, COUNT(*) as count
3 FROM
4   `still-kit-459403-e2.global._Default._Default`
5 GROUP BY resource.type

```

Results (1) Table Chart Both

Download Create alert Save to dashboard

Row	type	count
	STRING	INTEGER
1	gce_instance	10

Chart

Chart display

Chart type: Bar chart

Vertical Horizontal

Dimension (x-axis)

Column * type Limit * Top 5

Order default

Displaying top 5, sorted by first measure

Measure (y-axis)

Measure * max of count